THE LOCALIZATION OF A RING AT THE POINTS SPECIALIZING TO A LOCALLY CLOSED SUBSCHEME OF THE FORM D(F) CAP V(I)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the multiplicative subset through the ring of sections on Z, the two-sided characterization of the primes surviving in the localization, and the formal derivation of the dependence, the closed subscheme statement and the functoriality.

Editorial note. The label lemma-localization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-localization –start 709 –end 748 –local-id 3596 returned RESULT=OK (L3596 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of sections of the structure sheaf on a basic open of a closed subscheme. Algebra and Schemes, the identification of the global sections of the structure sheaf of D(f) cap V(I) with the localization of A/I at f. Supplies the description of the ring whose invertible elements define the multiplicative subset S.
- Localization and the spectrum of a multiplicative subset. Algebra, the standard description of the primes of a localization as the primes of A disjoint from the multiplicative subset. Supplies the two inclusions characterizing which primes lie in the spectrum of S^{-1}A.

Lemma 0.1. *Let A be a ring. Set $X = \mathrm{Spec}(A)$. Let $Z \subset X$ be a locally closed subscheme which is of the form $D(f) \cap V(I)$ for some $f \in A$ and ideal $I \subset A$. Then*

1. *there exists a multiplicative subset $S \subset A$ such that $\mathrm{Spec}(S^{-1}A)$ maps by a homeomorphism to the set of points of X specializing to Z,*
2. *the A-algebra $A_Z^\sim = S^{-1}A$ depends only on the underlying locally closed subset $Z \subset X$,*
3. *Z is a closed subscheme of $\mathrm{Spec}(A_Z^\sim)$,*

If $A \to A'$ is a ring map and $Z' \subset X' = \mathrm{Spec}(A')$ is a locally closed subscheme of the same form which maps into Z, then there is a unique A-algebra map $A_Z^\sim \to (A')_{Z'}^\sim$.

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Proof. Let $S \subset A$ be the multiplicative set of elements which map to invertible elements of $\Gamma(Z, \mathcal{O}_Z) = (A/I)_f$. If $\mathfrak{p}$ is a prime of A which does not specialize to Z, then $\mathfrak{p}$ generates the unit ideal in $(A/I)_f$. Hence we can write $f^n = g + h$ for some $n \geq 0$, $g \in \mathfrak{p}$, $h \in I$. Then $g \in S$ and we see that $\mathfrak{p}$ is not in the spectrum of $S^{-1}A$. Conversely, if $\mathfrak{p}$ does specialize to Z, say $\mathfrak{p} \subset \mathfrak{q} \supset I$ with $f \notin \mathfrak{q}$, then we see that $S^{-1}A$ maps to $A_\mathfrak{q}$ and hence $\mathfrak{p}$ is in the spectrum of $S^{-1}A$. This proves (1).

The isomorphism class of the localization $S^{-1}A$ depends only on the corresponding subset $\mathrm{Spec}(S^{-1}A) \subset \mathrm{Spec}(A)$, whence (2) holds. By construction $S^{-1}A$ maps surjectively onto $(A/I)_f$, hence (3). The final statement follows as the multiplicative subset $S' \subset A'$ corresponding to Z' contains the image of the multiplicative subset S. $\square$